Hitesh Patil

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Education

AISSMS Institute of Information Technology — Pune, India

B.Tech in Electronics and Telecommunication Engineering — *2025 – Present*

- GPA: 8.05 / 10

Experience

Team Lead – NIDAR 2027 RescueSwarm — Pune, India

Autonomous Multi-Drone Survivor Search & Aid Delivery Challenge — *Current*

- Leading a 10-member team building a collaborative autonomous multi-drone system for the NIDAR 2027 Track 1 Drone Innovation challenge, tasked with search, survivor geotagging, and aid delivery.
- Own the computer vision pipeline for real-time human/survivor detection using YOLO11n, deployed on an edge Raspberry Pi 5 for onboard, network-independent inference.
- Architecting the mission-critical Ground Control Station interface for unified multi-drone telemetry, mission planning, and live survivor/delivery status reporting to a single operator.

AI/ML Cohort Member — Pune, India

Catalyst Tech Club — *Current*

- Selected as part of the college AI/ML cohort focused on applied machine learning, EDA, and project-based learning.
- Attended technical sessions and contribute ∼40 hours/week toward self-learning and project development.
- Built applied ML and backend-integrated projects to reinforce concepts learned through the cohort.

Projects

WalkSafe | *Python, FastAPI, Flutter, Docker, Valhalla, OSMnx, PostGIS, H3, Shapely, ngrok* — GitHub

- Built an end-to-end geospatial navigation system that prioritizes pedestrian safety over shortest-path routing.
- Designed a safety scoring model using OSM features (lighting, sidewalks, road type, speed).
- Implemented a spatial data pipeline with OSMnx, PostGIS, and H3 indexing to store scored road segments and aggregate neighborhood-level safety zones.
- Integrated a FastAPI backend with a self-hosted Valhalla routing engine and a Flutter mobile app: route visualization, live navigation, safety overlays, SOS alerts, and incident reporting.
- Deployed a shareable demo APK via ngrok-tunneled backend for Android testing without cloud infrastructure.

Mission Overlay | *Node.js, Electron, Leaflet, MAVLink, UDP* — GitHub

- Built a unified multi-vehicle Ground Control Station for autonomous drone fleet coordination, interoperable with ArduPilot SITL, Mission Planner, and QGroundControl.
- Implemented live MAVLink telemetry ingestion over UDP with real-time GPS map tracking, flight instruments (artificial horizon, compass ribbon), and per-vehicle EKF health, battery, and GPS status monitoring.
- Engineered a one-click autonomous mission wizard: imports KML/GeoJSON search boundaries, auto-partitions the area across drones with feasibility checks, generates search grids, and uploads/verifies waypoints via MAVLink readback.
- Built in safety failsafes (RTL, geofencing, emergency recall) and packaged the app as a standalone Windows installer via Electron/NSIS.

Ignite 2.0 | *Next.js, TypeScript, Supabase, SQL, Tailwind, Framer Motion* — Live | GitHub

- Built a realtime tournament management platform with real-time score tracking, dynamic leaderboards, and a remote-controlled countdown/lock screen powered by Supabase Realtime.
- Engineered live standings with professional tie-breaking rules calculated on-the-fly from Postgres using `postgres_changes` subscriptions for millisecond-level updates.
- Implemented role-based auth (admin/scorer) with environment-driven credentials and a Supabase Auth provisioning script; scorer role limited to Fixtures and Scores tabs.
- Designed a "Dark Stadium" UI system using glassmorphism cards, Bebas Neue typography, and Framer Motion animations; achieved a cohesive premium aesthetic across all mobile viewports.

Technical Skills

Languages: Python, TypeScript
Frameworks: FastAPI, Streamlit, Next.js 14
Libraries: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Framer Motion, Tailwind CSS
Tools: Git, GitHub, Linux, Docker, ngrok, venv, PostgreSQL
Concepts: EDA, Feature Scaling, Hyperparameter Tuning, REST APIs, Geospatial Processing, Model Deployment, Realtime Databases, Role-Based Access Control

Additional Information

Comfortable with Git branching, merge requests, and PR-based collaborative workflows.
Familiar with Linux command-line usage, filesystem navigation, and light shell scripting.
Active GitHub profile at github.com/TheGoodDevil00 with ongoing project development.